

Orchestration and Scaling

Graded Project — StreamingApp Deployment

CI/CD with Jenkins, Amazon ECR, EKS, Helm, CloudWatch and SNS

GitHub - <https://github.com/kaushik0203/StreamingApp.git>

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```
streamingapp-frontend:latest 5018d5ee5a6a 78.2MB 21.9MB
streamingapp-streaming:latest 1b125f4189c8 302MB 74.7MB
ubuntu@ip-172-31-0-242:~/StreamingApp$ aws --version
aws-cli/2.35.11 Python/3.14.5 Linux/6.17.0-1017-aws exe/x86_64.ubuntu.24
ubuntu@ip-172-31-0-242:~/StreamingApp$
```

1. AWS CLI

```
ubuntu@ip-172-31-0-242:~/StreamingApp$ docker images
IMAGE ID DISK USAGE CONTENT SIZE EXTRA
712934020955.dkr.ecr.ap-south-1.amazonaws.com/streamingapp-admin:latest 6bf7c8adc7c1 271MB 59.5MB U
712934020955.dkr.ecr.ap-south-1.amazonaws.com/streamingapp-auth:latest d9723672e079 241MB 56.3MB U
712934020955.dkr.ecr.ap-south-1.amazonaws.com/streamingapp-chat:latest a70bd95e39a2 230MB 54.8MB U
712934020955.dkr.ecr.ap-south-1.amazonaws.com/streamingapp-frontend:latest 5018d5ee5a6a 78.2MB 21.9MB U
712934020955.dkr.ecr.ap-south-1.amazonaws.com/streamingapp-streaming:latest 1b125f4189c8 302MB 74.7MB U
mongo:6 8b6d8f5bbdb 1.1GB 282MB U
nginx:1.27-alpine 65645c7bb6a0 74.5MB 21.9MB U
node:19-alpine 8d6421d603a4 181MB 45.3MB U
streamingapp-admin:latest 6bf7c8adc7c1 271MB 59.5MB U
streamingapp-auth:latest d9723672e079 241MB 56.3MB U
streamingapp-chat:latest a70bd95e39a2 230MB 54.8MB U
streamingapp-frontend:latest 5018d5ee5a6a 78.2MB 21.9MB U
streamingapp-streaming:latest 1b125f4189c8 302MB 74.7MB U
ubuntu@ip-172-31-0-242:~/StreamingApp$
```

2. Docker Images

```
ubuntu@ip-172-31-0-242:~/StreamingApp$ kubectl get nodes
NAME STATUS ROLES AGE VERSION
ip-192-168-26-138.ap-south-1.compute.internal Ready <none> 24m v1.34.9-eks-93b80c6
ip-192-168-41-200.ap-south-1.compute.internal Ready <none> 24m v1.34.9-eks-93b80c6
ubuntu@ip-172-31-0-242:~/StreamingApp$
```

3.EKS Nodes

```
Username for 'https://github.com': kaushik0203
Password for 'https://kaushik0203@github.com':
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 2 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (7/7), 1.82 KiB | 620.00 KiB/s, done.
Total 7 (delta 4), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (4/4), completed with 2 local objects.
To https://github.com/kaushik0203/StreamingApp.git
 4e433cf..d89233a main -> main
ubuntu@ip-172-31-0-242:~/StreamingApp$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
ubuntu@ip-172-31-0-242:~/StreamingApp$ git log --oneline -5
d89233a (HEAD -> main, origin/main, origin/HEAD) Add Jenkins pipeline and ignore local binaries
4804d58 Add Jenkins pipeline and ignore local binaries
4e433cf Deploy StreamingApp on AWS EKS using Kubernetes manifests
070285d Update README.md
3619f6b WIP: bugs fixed on chat and streaming, forget password, others as per the ticket #492SFL
ubuntu@ip-172-31-0-242:~/StreamingApp$
```

4. GIT Log Last

```

ubuntu@ip-172-31-0-242:~/StreamingApp$ kubectl get pods -n streamingapp
NAME                                READY    STATUS    RESTARTS   AGE
admin-6c6544d6f8-67dtz             1/1      Running   0           4m38s
auth-75b797bf-t4wmm                1/1      Running   0           4m46s
chat-6d7c49f97-dl9rl               1/1      Running   0           4m35s
frontend-7c87bfd96b-bbnhp          1/1      Running   0           4m32s
mongo-7c94fcd666-9lv2z             1/1      Running   0           4m51s
streaming-6fb8954c7-rqqrj          1/1      Running   3 (4m46s ago) 6m36s
ubuntu@ip-172-31-0-242:~/StreamingApp$ kubectl get svc -n streamingapp
NAME                                TYPE                CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
admin-service                       ClusterIP            10.100.190.244   <none>            3003/TCP          5m19s
auth-service                        ClusterIP            10.100.170.234   <none>            3001/TCP          5m27s
chat-service                        ClusterIP            10.100.114.76    <none>            3004/TCP          5m16s
frontend-service                    LoadBalancer        10.100.106.35    a446f6bd1f30747d690bd222b17407d8-1511197498.ap-south-1.elb.amazonaws.com 80:30585/TCP      5m13s
mongo                               ClusterIP            10.100.59.189    <none>            27017/TCP         5m32s
streaming-service                   ClusterIP            10.100.1.146     <none>            3002/TCP          7m17s
ubuntu@ip-172-31-0-242:~/StreamingApp$

```

5. Running Pods

```

ubuntu@ip-172-31-0-242:~/StreamingApp$ kubectl get pods -n streamingapp
NAME                                READY    STATUS    RESTARTS   AGE
admin-6c6544d6f8-67dtz             1/1      Running   0           4m38s
auth-75b797bf-t4wmm                1/1      Running   0           4m46s
chat-6d7c49f97-dl9rl               1/1      Running   0           4m35s
frontend-7c87bfd96b-bbnhp          1/1      Running   0           4m32s
mongo-7c94fcd666-9lv2z             1/1      Running   0           4m51s
streaming-6fb8954c7-rqqrj          1/1      Running   3 (4m46s ago) 6m36s
ubuntu@ip-172-31-0-242:~/StreamingApp$

```

6. Running Pods

```

ubuntu@ip-172-31-0-242:~/StreamingApp$ kubectl get svc -n streamingapp
NAME                                TYPE                CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
admin-service                       ClusterIP            10.100.190.244   <none>            3003/TCP          7m12s
auth-service                        ClusterIP            10.100.170.234   <none>            3001/TCP          7m20s
chat-service                        ClusterIP            10.100.114.76    <none>            3004/TCP          7m9s
frontend-service                    LoadBalancer        10.100.106.35    a446f6bd1f30747d690bd222b17407d8-1511197498.ap-south-1.elb.amazonaws.com 80:30585/TCP      7m6s
mongo                               ClusterIP            10.100.59.189    <none>            27017/TCP         7m25s
streaming-service                   ClusterIP            10.100.1.146     <none>            3002/TCP          9m10s
ubuntu@ip-172-31-0-242:~/StreamingApp$

```

7. Services + load Balancer